



Enhancing volatile fatty acid (VFA) and bio-methane production from lawn grass with pretreatment



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HIGHLIGHTS

- Lawn grass were pretreated by four mild methods to reduce cost of AD process.
- The relation between VFA and bio-methane was decoupled to study VFA production.
- High ozone dose increased VFA production and decreased bio-methane production.
- SAA was the most effective pretreatment for improving yields of both VFA and bio-methane.

ARTICLE INFO

Article history:

Received 9 January 2014
Received in revised form 15 March 2014
Accepted 18 March 2014
Available online 26 March 2014

Keywords:

Pretreatment
Anaerobic digestion
Volatile fatty acid (VFA)
Bio-methane

ABSTRACT

The bioconversion of fiber-based carbohydrates during anaerobic digestion (AD) is impeded due to the recalcitrant nature of the plant cell wall. Pretreatment of lignocellulose materials under mild conditions are needed to improve the digestibility at minimum cost. This study investigated the effects of different pretreatments, including ozone, soaking aqueous ammonia (SAA), combined ozone and SAA (OSAA), and size reduction to enhance volatile fatty acid (VFA) and bio-methane production when lawn grass was used as substrate. To study VFA production, methanogenesis was selectively inhibited by sodium 2-bromoethanesulfonate to decouple the relation between VFA and bio-methane. The enzymatic hydrolysis of SAA (residence time 24 h at 50 °C) and OSAA (10 min ozonation and 6 h of SAA) in pretreatment of lawn grass sample resulted in 86.71% and 89.63% sugar recovery, respectively. The specific methane yields of the control, ozone, SAA, OSAA, and size-reduced grass samples were 402.5, 358.8, 481.0, 462.6, and 358.3 ml CH₄/g volatile solid (VS), respectively.

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1. Introduction

Anaerobic digestion (AD) for converting bio-wastes to biogas has the potential to replace fossil fuels and reduce environmental impact (Yu et al., 2012). Grass as an abundant renewable feedstock has enormous significant potential to for this application. For example, the available estimate of turf grass in the USA is 27.6 million acres, while residential lawn is about 21 million acres (Bormann et al., 2001). In Europe, especially in Germany, grass silage is considered an important feedstock or co-substrate for AD plants (Nizami and Murphy, 2010).

One of the main obstacles in utilizing grass biomass as feedstock for biogas production is its complex structure. The recalcitrant components present in grass biomass and their chemical bonding association with each other limit access of microorganisms and

enzymes for efficient digestion (Nizami et al., 2009; Steenari et al., 2009). The most recalcitrant part of the biomass is lignin due to its hydrophobic nature (Lehtomaki et al., 2008). Inefficient biodegradation results in reduced solubility of the grass biomass components, limiting conversion of total chemical oxygen demand (COD) to soluble COD. Enhanced biodegradation and solubility of crystalline cellulose and hemicellulose could lead to more efficient hydrolysis of grass biomass. As biodegradation of structural carbohydrates in AD is a rate limiting process, requiring a large volume of digester and a longer residence time, the resultant overall cost (construction and operational) of the system increases. This imposes the need to improve overall efficiency of the AD process in a biogas plant.

The hydrolysis rate can be accelerated by pretreating the substrate with chemical, physical, and biological processes before feeding to the digester (Sreekrishnan et al., 2004). Pretreatment decreases the structural and compositional barriers of grass biomass to digestion by breaking down lignin interference, reducing cellu-

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lose crystallinity, and increasing surface area. This ultimately enhances the efficiency of methane production (Mosier et al., 2005). Among different pretreatment technologies NaOH was widely recognized in the pretreatment of lignocellulosic biomass for higher hydrolysis yield and biogas productivity (Frigon et al., 2012; Liew et al., 2011; Xie et al., 2011). Work by Xie et al. (2011) showed that the solubility of total COD was increased by 45%, with removal of lignin, hemicellulose and cellulose at 65.6%, 36.1% and 21.2%, respectively; the highest methane yield was 452.5 ml CH₄/g VS at 5% NaOH and temperature 100 °C. However, NaOH can cause corrosion damage of the reactor, especially at high temperature, and can also increase the risk of degradation and loss of carbohydrates (Zuo et al., 2012).

Soaking aqueous ammonia (SAA) at low temperature and pressure appears to be an attractive pretreatment method for AD (Himmelsbach et al., 2010). Aqueous ammonia has been demonstrated effective in the treatment of low lignin feedstocks, such as agriculture residues. One of the important features of ammonia-based pretreatment is that most of the hemicelluloses are retained after treatment, which enhances the conversion efficiency (Zuo et al., 2012). Ozonation is traditionally used for wastewater in advanced oxidation processes, as well as for reduction of sewage sludge in wastewater treatment plants (Cesaro and Belgiorno, 2013). Ozone pretreatment effectively removes lignin and is carried out at room temperature and pressure (Sun and Cheng, 2002). Therefore, combining ozone and SAA (OSAA) is a promising treatment method to degrade lignocellulosic biomass under mild conditions. Since the previous researches on ozone and SAA pretreatment only provided the results of biomethane production, the effects on volatile fatty acids (VFA) are still poorly understood. Furthermore, studies are limited on the use of pretreatments to produce VFA by AD, which could provide value-added products. In the typical composition of VFA, the ratio of acetate:propionate:butyrate is 6:1:3 or 7:1:2 (Chang et al., 2010). Increasing propionate productivity will add potential benefits to the AD process. For example, propionic acid is a very important chemical widely used in the production of cellulose plastics, herbicides, perfumes, food, and feed preservative (Dishisha et al., 2012). Commercial production of propionic acid is predominantly carried out via petrochemical routes (Liang et al., 2012). To compensate for the decline of oil resources, biological routes allow producing propionic acid from organic wastes, such as green wastes. In addition, efficient and economical pretreatment methods for AD are still in demand for enhancing bio-methane productivity.

In this study, four pretreatment methods at mild temperature (≤ 50 °C) were studied to enhance VFA and bio-methane productivity, considering the grass waste as potential feedstock for AD processing. To investigate VFA production, sodium 2-bromoethanesulfonate was chosen to selectively inhibit methanogen instead of other bacteria (Xu et al., 2010; Zinder et al., 1984). Different pretreatment methods, such as ozone, SAA, OSAA, and size reduction, were verified to determine their effect on VFA and bio-methane productivity. In addition, the effects of various types of biomass feedstocks, methanogenic inhibition, total solid (TS) and VS on anaerobic biodegradation were investigated.

2. Methods

2.1. Feedstock and inoculums

Residential grass clippings were collected in Pullman, WA. *Poa pratensis* L., the scientific name of the grass, is commonly known as Kentucky Bluegrass, a perennial species of grass native to North America, Europe, Asia, and northern Africa. The grass was collected from lawnmower clippings and dried at 60 °C. After drying for 24 h,

the grass waste was ground to a particle size of less than 1 mm in a hammer mill. The grass samples were then sieved through a 32–42 mesh (Tyler Standard Screen Scale). Thereafter, grass waste particles of 0.354–0.5 mm were selected to conduct the different experiments. Anaerobic sludge containing a microbial community of hydrolyzing, acid-producing, acetate-producing, and methane-producing microbes was sampled from an anaerobic digester located at the Pullman Wastewater Treatment Facility. The sludge with a concentration of 17.1 g TS/L (11.7 g VS/L) and pH of 7.5 was used as inoculum for anaerobic tests. Table 1 shows the characteristics of grass waste and inoculum used during the study.

2.2. Batch anaerobic digestion

A substrate/inoculum ratio of 1.0 g VS waste/g VS inoculum was selected to compare the VFA and biomethane production before and after pretreatment. All of the experiments were performed in triplicate for different waste materials and conditions studied. Batch experiments for AD of grass waste were conducted in 250 ml glass serum bottles with a liquid volume of 120 ml, at 37 ± 1 °C controlled by a thermostat of the respirometry system. These experiments were running in a thermostatic room controlled by an electric air heater. Before the anaerobic experiments, the glass serum bottles with mixed liquor were put in an anaerobic chamber (BACTRON™ Anaerobic/Environmental Chamber, SHEL LAB) to remove oxygen using vacuum and nitrogen. The anaerobic bottle used a 0.6 mm needle to release biogas for analysis. The following different procedures were used to study VFA and biomethane production, respectively: 10 mM Sodium 2-bromoethanesulfonate (C₂H₄BrNaO₃S, Sigma–Aldrich Corporation, USA) was used to inhibit methanogenic bacteria for the VFA accumulation (Chidthaisong and Conrad, 2000). The sampling for VFA analysis was operated in the anaerobic chamber, and the analytical method was described in Section 2.5.2. The biochemical methane potential (BMP) assay was undertaken to evaluate biomethane production (Ma et al., 2013). Methane production was monitored using a respirometer (Challenge technology AER-200, Springdale, AR). Sodium hydroxide was used as scrubbing media to purify methane by adsorbing carbon dioxide and hydrogen sulfide, so that only methane production was recorded.

2.3. Pretreatment

2.3.1. Ozone pretreatment procedure

The resulting 0.354–0.5 mm grass particles were used for the ozone pretreatment. Ozone was produced by a L11–L24 Ozone Generator manufactured by Pacific Ozone, CA, USA. The grass par-

Table 1
Characteristics of grass waste and inoculums.

Parameters	Unit	Value
<i>Grass waste particles</i>		
Total solid (TS)	% (w/w)	96.27 $\pm$ 0.04
Total volatile solid (TVS)	% (w/w)	68.09 $\pm$ 0.14
Ash	% (w/w)	27.15 $\pm$ 1.27
<i>Active sludge</i>		
Total solid (TS)	% (w/w)	1.54 $\pm$ 0.002
Total volatile solid (TVS)	% (w/w)	1.18 $\pm$ 0.003
Ash	% (w/w)	0.36 $\pm$ 0.001
<i>Feedstock</i>		
Total solid (TS)	% (w/w)	2.54 $\pm$ 0.004
Total volatile solid (TVS)	% (w/w)	1.73 $\pm$ 0.001
Ash	% (w/w)	0.81 $\pm$ 0.003
Total COD	g/L	222.65 $\pm$ 55.35
Total nitrogen	g/L	0.84 $\pm$ 0.03
Total phosphorus	g/L	0.27 $\pm$ 0.02

ticles (3 g) were adjusted for moisture (90% w/w) and placed into an enclosed stainless steel reactor, with 1 cm bed height to ensure equal contact time between all particles with ozone and operated in a semi-continuous mode. The ozonation reaction was performed under 5.3% ozone concentration (w/w) at a flow rate of 2000 ml/min for 10 min (Bule et al., 2013). Ozone dose was 0.79 g O₃/g TS. These processed samples were used for further investigation.

2.3.2. Soaking aqueous ammonia (SAA) pretreatment procedure

The grass waste particles were pretreated by soaking aqueous ammonia (SAA) using 28–30% (w/w) ammonium hydroxide solution (JTB-9721-03), with a solid to liquid ratio of 2:10 at 50 °C for 24 h without agitation in the 1000 ml screw cap Pyrex solution bottle (Gao et al., 2012). After completion of incubation, the grass was washed twice thoroughly with distilled water to reach a neutral pH. The obtained cake was dried at 50 °C and used for further investigation.

2.3.3. Ozone combined with SAA (OSAA) pretreatment procedure

Grass waste (32–42 mesh) was pretreated by the aforementioned ozone pretreatment (Section 2.3.1) for 10 min and followed by SAA pretreatment (Section 2.3.2) for 6 h. The resulting material was used for further investigation.

2.3.4. Size reduction

The milled grass waste particles were sieved through a 42 mesh Tyler Standard Screen Scale to obtain the grass waste particles with a size less than 0.354 mm.

2.4. Enzymatic hydrolysis

The enzymatic hydrolysis of pretreated lawn grass was performed at 1% (w/v) solid loading in 0.05 M sodium citrate buffer (pH 4.8) containing 100 µl 2% sodium azide with 30 FPU/g of cellulase (Novozymes NS 50013) and 30 CBU/g of β-glycosidase (Novozymes NS 50010). The enzymatic reaction was performed at 50 °C in an orbital incubator shaker (Gyromax 747) for 72 h. The total released sugars after 72 h of incubation were measured using DIO-NEX-IC to determine sugar recovery.

2.5. Analytical methods

2.5.1. Biomass composition, acid soluble and insoluble lignin analysis

Control and pretreated lawn grass samples were dried at 105 °C in hot air oven for 24 h and then used for compositional analyses. Carbohydrate and lignin content were determined according to the National Renewable Energy Laboratory (NREL)'s standard laboratory analytical procedure (LAP) for determination of structural carbohydrates in biomass (Sluiter et al., 2008). A 0.3 g sample was weighed and used in a two-stage acid hydrolysis method. After initial hydrolysis at 37 °C with 3 ml of 72% (w/w) sulfuric acid, the samples were diluted with distilled water to a total volume of 84 ml and autoclaved for 1 h in pressure tubes. Sugars in the aqueous phase were determined using ion chromatography (Dionex ICS-3000 with Dionex Pac PA20 column and CarboPac PA20 guard column). Samples were run for 60 min and the column was flushed between runs with 100% 200 mM NaOH followed by de-ionized water. Sugar concentration was calculated by comparison to a standard sugar sample, and all measurements were taken in triplicate. Acid soluble and insoluble lignin contents were calculated according to NREL's standard laboratory analytical procedure (Sluiter et al., 2005, 2008), respectively.

2.5.2. Volatile fatty acid (VFA) analysis

The concentrations in the effluent of VFA, including acetate, propionate, butyrate, i-butyrate, valerate, and caporate, were

determined daily with a gas chromatograph (Shimadzu Corp., Japan, Model GC-2014) equipped with a flame ionization detector and length of 30 m × id 0.25 mm × film thickness 0.25 µm capillary column (HP-INNOWax, Agilent Technologies, Palo Alto, CA). The liquor samples were first centrifuged at 12,000 rpm for 5 min, then acidified with formic acid and filtrated through 0.22 µm membrane, and finally measured for free acids. The temperatures of the injector and detector were 250 °C and 300 °C, respectively. Initial oven temperature and time was 70 °C for 3 min followed by an increase of 15 °C/min to the final temperature of 230 °C for 3 min. Nitrogen was used as carrier gas with a flow rate of 0.93 ml/min.

2.5.3. Other analysis

Total chemical oxygen demand (COD), total phosphorus (TP), total solids (TS), and volatile solids (VS) were determined according to standard methods (Eaton et al., 2005). The pH was measured daily using a Fisher Scientific AB15 pH meter. Total Kjeldahl nitrogen (TKN) was measured using a 2300 Kjeltac Analyzer Unit (Tecator, Perstorp Analytical).

3. Results and discussion

3.1. Effect of different pretreatments on grass waste composition

Table 2 shows the effects of the different pretreatments on the composition of grass waste samples. It was observed that sugar concentration in the grass waste particles decreased slightly from 30.55% (unpretreated) to 28.58% (ozone pretreated) after 10 min ozone pretreatment, which was likely due to oxidation of sugar components. Ozone is known to be selective in lignin degradation during the pretreatment process, and degrades a minimum amount of cellulose (Sun and Cheng, 2002). Ben-Ghedalia and Miron (1981) and Binder et al. (1980) reported loss of cellulose after ozone pretreatment as below 5% when working with low pH values. Overall sugar content with SAA and OSAA pretreatment was increased by 46.55% and 46.73%, respectively. The increased sugar concentration could be due to removal of ash, extractives, and conversion of insoluble lignin into soluble lignin. The acid soluble lignin content of grass waste increased from 3.73% to 4.18% after ozone pretreatment. The insoluble lignin content decreased likewise from 16.66% to 14.21% after ozone pretreatment. With SAA pretreatment the soluble lignin content was 1.89%, while acid insoluble lignin content was 13.41%. For OSAA pretreatment, the lowest acid insoluble lignin content was observed as 11.13%, while soluble lignin content was 2.51%. The ash value decreased from

Table 2
Compositional analysis of unpretreated and pretreated grass.

Components	Control grass (% w/w)	Ozone (% w/w)	SAA (% w/w)	OSAA (% w/w)
<i>Sugars</i>				
Arabinose	2.90 ± 0.08	1.90 ± 0.01	3.70 ± 0.01	4.35 ± 0.36
Galactose	1.23 ± 0.02	0.89 ± 0.01	1.28 ± 0.01	0.83 ± 0.02
Glucose	16.91 ± 1.07	16.14 ± 0.11	30.85 ± 0.02	30.96 ± 0.13
Xylose	9.52 ± 0.50	9.65 ± 0.06	10.73 ± 0.04	10.59 ± 0.04
Total	30.55 ± 0.46	28.58 ± 0.20	46.55 ± 0.05	46.73 ± 0.54
<i>Lignin</i>				
Acid insoluble	16.66 ± 0.57	14.21 ± 0.13	13.41 ± 0.86	11.13 ± 1.01
Acid soluble	3.73 ± 0.09	4.81 ± 0.72	1.89 ± 0.34	2.51 ± 0.47
<i>Extractives</i>				
Ethanol	5.83 ± 0.21	ND	ND	ND
Water	4.18 ± 0.39	ND	ND	ND
Total proteins	1.52 ± 0.002	ND	ND	ND
Ash	27.15 ± 1.27	26.43 ± 0.74	13.00 ± 0.38	11.29 ± 0.51

ND: not determined.

Table 3
Sugar recovery of lawn grass sample after different pretreatments.

Sample	Sugar recovery (% w/w)
Unpretreated grass	0.00 ± 0.00
Ozone pretreated grass	48.50 ± 2.17
SAA pretreated grass	86.71 ± 0.20
OSAA pretreated grass	89.63 ± 2.09

27.15% to 13% and 11.29% with SAA and OSAA pretreatments, respectively, while minor decrease was reported in the ozone-pretreated sample (26.43%).

3.2. Effect of grass pretreatment on sugar recovery

Table 3 shows the sugar recovery from the different grass samples with and without pretreatment. It was observed that cellulosytic enzymes were unable to hydrolyze sugars from unpretreated grass samples. Ozone pretreatment for 10 min improved sugar recovery up to 48.50%. This yield is comparable to previous reports on wheat straw where 2.5 h long pretreatment resulted in 88.6% sugar recovery (García-Cubero et al., 2009). SAA pretreatment of grass sample for 24 h at 50 °C resulted in 86.71% sugar recovery. Although sugar recovery was improved, long process time puts SAA at a disadvantage; hence, a combination pretreatment of ozone and SAA was tested. The purpose of OSAA pretreatment was to use ozone to specifically deconstruct lignin and use SAA to change the crystallinity of cellulose, which should ultimately improve enzymatic hydrolysis. The combination pretreatment of ozone for 10 min and SAA for 6 h resulted in 89.63% sugar recovery, and therefore minimizes the cost barrier of ozone and long process time of SAA.

3.3. Effect of grass pretreatment on bio-methane production

Fig. 1 illustrates bio-methane the production profile over 30 days with grass samples pretreated using different methods. After the 3 day lag phase, a significant increase of bio-methane generation was observed in the initial days of digestion; thereafter the methane production slowed to almost zero. Of the total bio-methane produced, more than 99% was produced in about 10 days of digestion. As the grass wastes with low VS (1.73%) were fed to the digesters, pH was maintained at about 7.5. The samples were tested every two days, during which time no VFA was detected in the digesters.

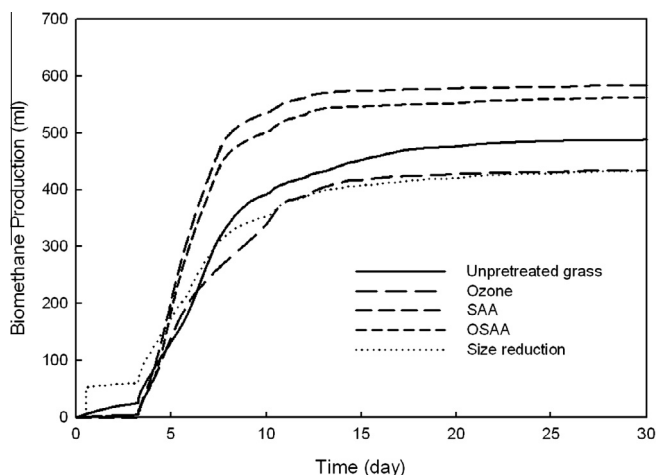


Fig. 1. Changes in bio-methane over time with different pretreatment (unpretreated grass, ozone, SAA, OSAA and size reduction).

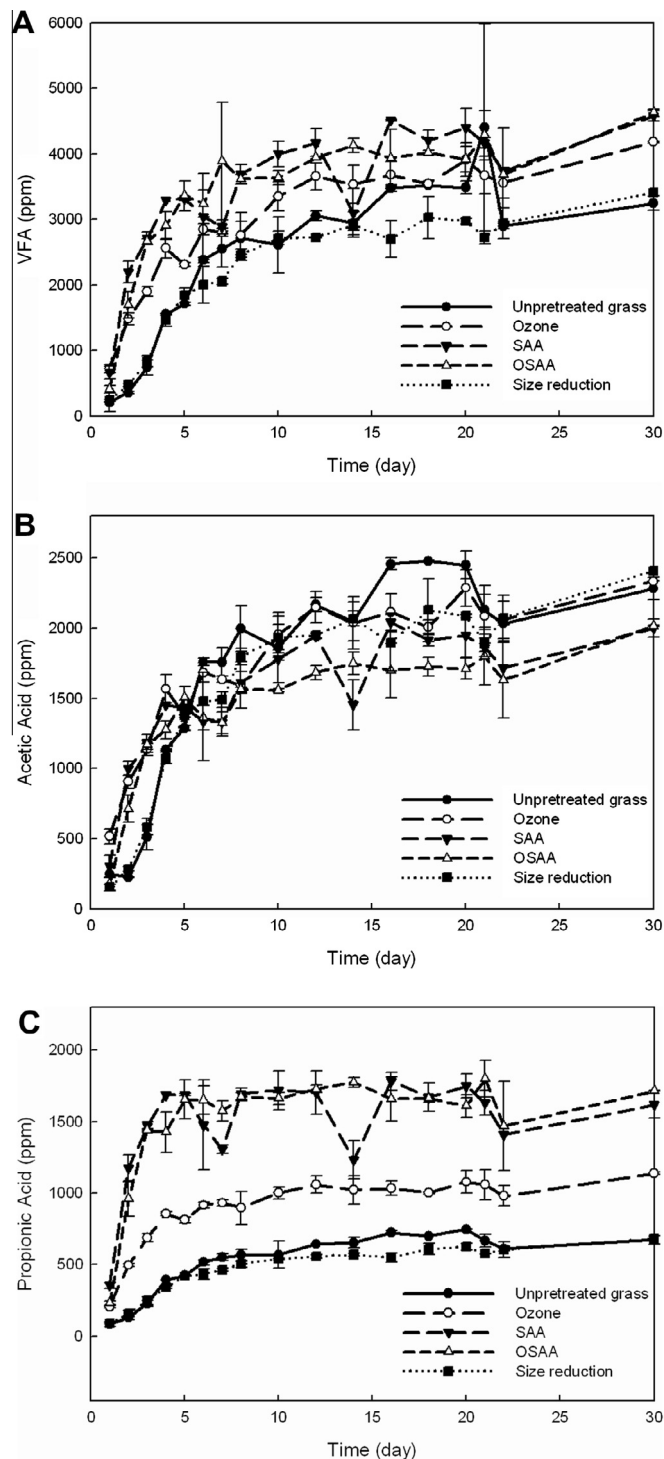


Fig. 2. Changes in (A) VFA, (B) acetic acid, (C) propionic acid with time with different pretreatment.

As shown in Fig. 1, the highest bio-methane production was observed in grass samples undergoing SAA pretreatment, with the highest specific methane yield at 481.0 ml CH₄/g VS. These results showed approximately a 30% increase in bio-methane production within 10 days of incubation as compared to the unpretreated grass sample. Over 30 days of digestion, an increase of around 20% was observed comparing methane yields of SAA pretreated and unpretreated grass samples. The specific methane yields of unpretreated, ozone, OSAA, and size reduction pretreated samples were 402.5, 358.8, 462.6, and 358.3 ml CH₄/g VS, respectively.

The ozone pretreatment did not improved bio-methane production, although the enzymatic hydrolysis showed improved sugar recovery (see Table 2). This could be due to an overdose of ozone delivered during pretreatment. An earlier report suggested that 0.15 g O₃/g TS is the maximum limit for inhibition of AD by ozone (Bougrier et al., 2006). The ozone dose of 0.79 g O₃/g TS used in this study was considerably higher than the aforementioned limit. The bio-methane production in the pretreated grass sample using ozone decreased instead of increasing, which is in consistent with another previous report (Cesaro and Belgiorno, 2013). Bougrier et al. (2006) explained various reasons for the reduction of biogas production during ozone pretreatments. The formation of refractory compounds could be a more relevant explanation for this study. Methanogenic bacteria are more sensitive to changes than other microorganisms present in anaerobic digestion. This is due to the faster growth rate of the other groups, such as the acetogens, which can achieve substantial catabolism. The refractory compounds might be selectively inhibitory to methanogenic bacteria, as will be discussed in Section 3.4, which will show the positive effect of ozone pretreatment on VFA production. This explanation also could be confirmed from the results of OSAA pretreatment. Although OSAA pretreatment resulted in the highest sugar recovery among tested pretreatments in this study, it did not support the highest methane production. Methane production with SAA pretreatment was higher than with OSAA pretreatment by 5% over 10 days and 4% over 30 days.

The physical pretreatment (size reduction) also did not improve bio-methane production in grass waste. When bio-methane production results were compared in size-reduced and unpretreated grass samples, it was observed that bio-methane production decreased by 7% over 10 days and 12% over 30 days in the

size-reduced sample. The probable reason for decrease could be that the samples undergoing size reduction had higher ash content (33.66%) as compared to 27.15% in the unpretreated grass. The resulting decrease in bio-methane production is also related to the VFA production described in the next subsection (3.4).

3.4. Effect of grass pretreatment on VFA production

Fig. 2 describes the effects of different pretreatments on VFA accumulation over 30 days of fermentation. Similar to bio-methane production, a noteworthy increase of VFA concentration was observed in the first 7 days; after the VFA production slowed to almost zero. These experiments were conducted under the same conditions described in Section 3.3. In this study, to promote VFA accumulation, 10 mM sodium 2-bromoethanesulfonate was used to inhibit methanogenic bacteria instead of acetogenesis, as reported elsewhere (Xu et al., 2010). Less than 10% biogas was observed in the anaerobic process with methanogenic inhibition than that without methanogenic inhibition.

Fig. 2A describes the effect of grass waste pretreatment on the total VFA concentration. The results obtained with and without pretreatments show that VFA production was significantly increased in the SAA and OSAA pretreated samples. The statistical analysis of VFA production in the SAA and OSAA pretreated samples does not show a significant difference ($P > 0.05$). VFA production in the SAA pretreated sample was higher compared to the unpretreated sample by 53% over 10 days and 41% over 30 days. This indicates that the effect of pretreatment on VFA production decreased over time. Unlike the effect of ozone pretreatment on biomethane production, the VFA production was enhanced by

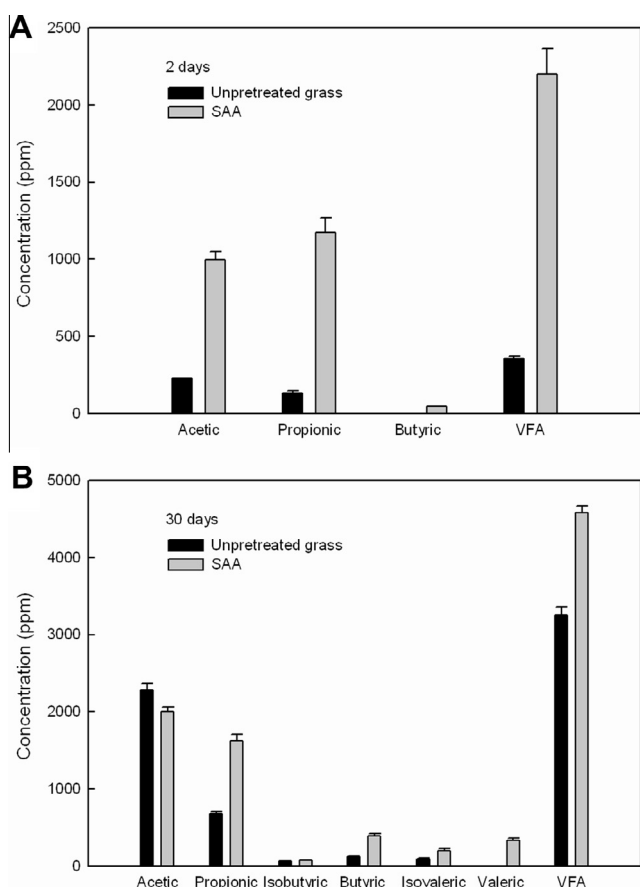


Fig. 3. Comparison of VFA production with and without pretreatment: (A) After 2 days, (B) after 30 days.

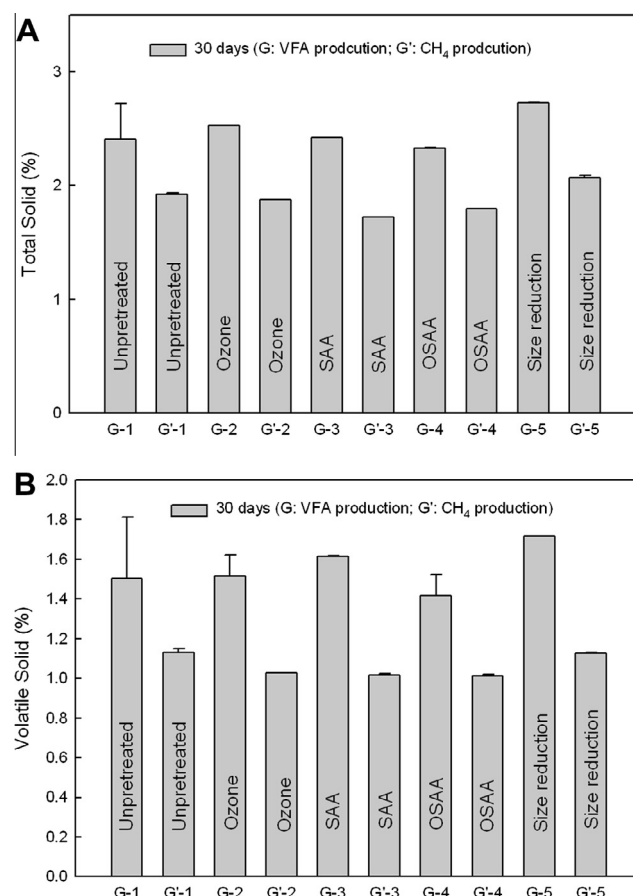


Fig. 4. Comparison of (A) TS, (B) VS reductions during VFA and bio-methane production.

28%, which indicates the ozone pretreatment did not inhibit acidogenic and acetogenic bacteria.

One-way ANOVA analysis also showed no significant difference ($P > 0.05$) between the size reduction and unpretreated grass waste, which indicates that the smaller particles did not improve VFA production in this study. These results corroborated the earlier findings in Fig. 1. It has been reported that biomass particle size reduction below 40 mesh (0.400 mm) has little effect on biomass hydrolysis and biogas production (Agbor et al., 2011; Nallathambi Gunaseelan, 1997).

Further, detailed analysis of the production profile of acetic acid and propionic acid are shown in Fig. 2B and C, respectively. A significant increase of 100–200% was observed in propionic acid production in pretreated grass waste with ozone, SAA, and OSAA. In contrast, acetic acid production was not prominently affected by ozone, SAA and OSAA pretreatment. Propionic acid, which is mainly supplied via petrochemical routes, is widely used in the production of cellulose plastics, herbicides, perfumes, food, and feed preservative (Dishisha et al., 2012). Based on the current results, it is possible to enhance propionic acid production significantly in the AD process using SAA and OSAA pretreatment.

Xie et al. (2011) suggested that the HPr (propionic acid) to HAC (acetic acid) ratio can be used as an indicator of digester imbalance. Hill et al. (1987) has proposed that an HPr to HAC ratio greater than 1.4 could indicate impending digester failure. In this study the HPr to HAC ratio was less than 1.0, which explains why the anaerobic process was maintained without failure or souring.

3.5. Comparison of VFA products before and after pretreatment

As SAA pretreatment proved to be the most effective in this study, a comparison of carboxylic acid concentration in the pretreated grass sample using SAA and the unpretreated grass waste was performed for 2 days and 30 days, as shown in Fig. 3A and B, respectively. A significant increase in carboxylic acid concentration was observed with SAA pretreatment after 2 days. The difference in carboxylic acid concentration between SAA pretreated and unpretreated grass waste became minimal over time. After day 30, the acetic acid concentration in the unpretreated grass waste (2282.32 ppm) was higher compared to the SAA pretreated sample (2000.16 ppm). The different types of carboxylic acid produced over time ranged from low molecular to high molecular weight. Therefore, it can be inferred that the short fermentation time is suitable for showing the effectiveness of SAA pretreatment in producing bio-methane, and if the purpose is to produce different types of carboxylic acid, a long fermentation time is required.

3.6. Comparison of TS/VS reductions in VFA and bio-methane production

Fig. 4A and B show the effects of different pretreatments on TS and VS after 30 days during VFA and bio-methane production. Under the aforementioned operational conditions, the VFA was produced by a methanogen inhibition mechanism, while bio-methane was produced without methanogen inhibition. The

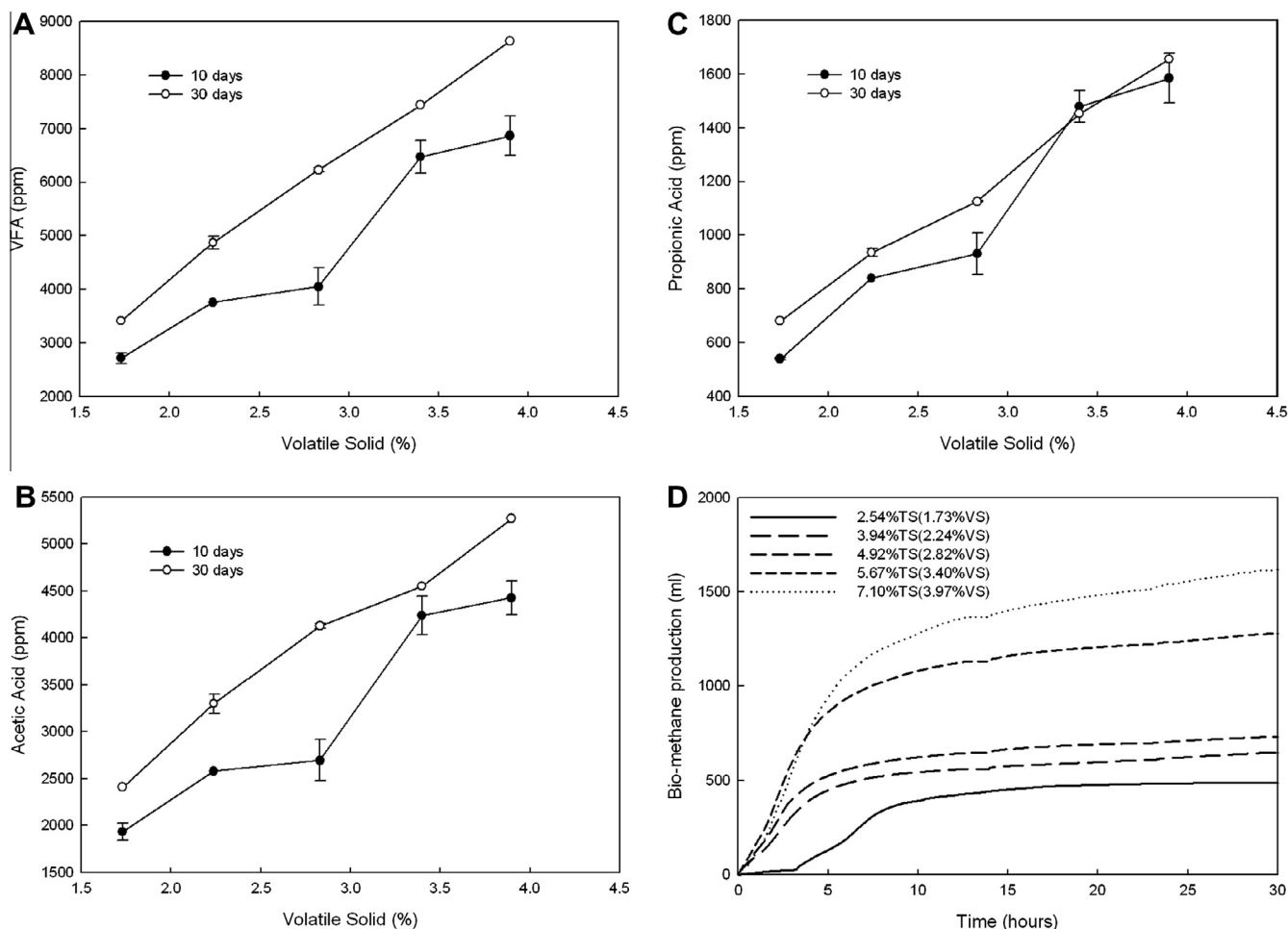


Fig. 5. Effect of VS on VFA and bio-methane production: (A) changes in VFA with VS, (B) changes in acetic acid with VS, (C) changes in propionic acid with VS, (D) changes in bio-methane over time.

results showed that more TS/VS reduction occurred in the bio-methane production as compared to VFA production. These results indicate that the methanogen inhibition would decrease TS/VS reductions significantly.

3.7. Effect of VS on VFA production and bio-methane production

Fig. 5A–C shows the effect of VS on VFA, acetic acid and propionic acid, respectively. In this study, grass samples were used without any pretreatment during the AD process and 10 mM sodium 2-bromoethanesulfonate was used to inhibit methanogenic bacteria. A linear increase with an increase in VS was observed after 30 days. A steep increase was found in VS of 2.82 to 3.40 at 10 days, which indicated that there is a critical value for VS over which VFA production can be significantly increased. The results compared in Figs. 2A and 5A show that the same increased VFA production is observed in SAA pretreatment and untreated sample with 30% additional VS.

Fig. 5D shows that the bio-methane accumulation increased over time for the first 10 days of digestion and then gradually went flat. In this study, no pretreatment was used and methanogenic bacteria were not inhibited. Corresponding to the VFA production shown in Fig. 5, a considerable increase of bio-methane production was observed in the VS (2.82–3.40). These results support the aforementioned inference that a critical value exists for TS/VS wherein VFA production can be significantly increased, and therefore bio-methane production. The specific methane yields of grass waste decreased with increase of TS/VS except the range between 4.92%TS/2.82%VS and 5.67%TS/3.40%VS. The specific methane yields of grass waste with 2.54%TS, 3.94%TS, 4.92%TS, 5.67%TS and 7.10%TS were 402.5, 261.8, 194.3, 250.6, and 249.6 ml CH₄/g VS, respectively.

4. Conclusions

The VFA and bio-methane production using AD of lawn grass was enhanced by different pretreatments at mild temperature. SAA was found to be the most effective pretreatment for improving yields of both VFA and bio-methane, compared to the other pretreatments used in this study. Although ozone pretreatment decreased bio-methane productivity in this research, it increased VFA production. Therefore, while the sugar and VFA productivity of OSAA was not lower than that of SAA, the bio-methane productivity was reduced. This study suggests that pretreatment technologies with higher hydrolysis of biomass may not be able to enhance methane productivity.

Acknowledgements

The authors thank the Washington State Department of Ecology Waste to Fuels program and their funding via contract #C1000172, and also thank Cynthia Alwine for assisting in experimental analysis.

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